

ECE595: Power Converters for AC Systems

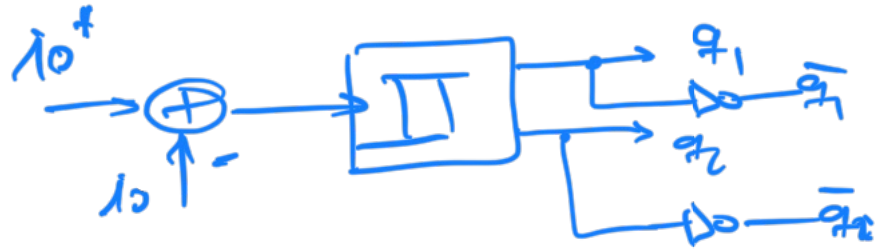
Lecture 23 *(Control for Power Converters – cont...)*

Prof. Euzeli Dos Santos

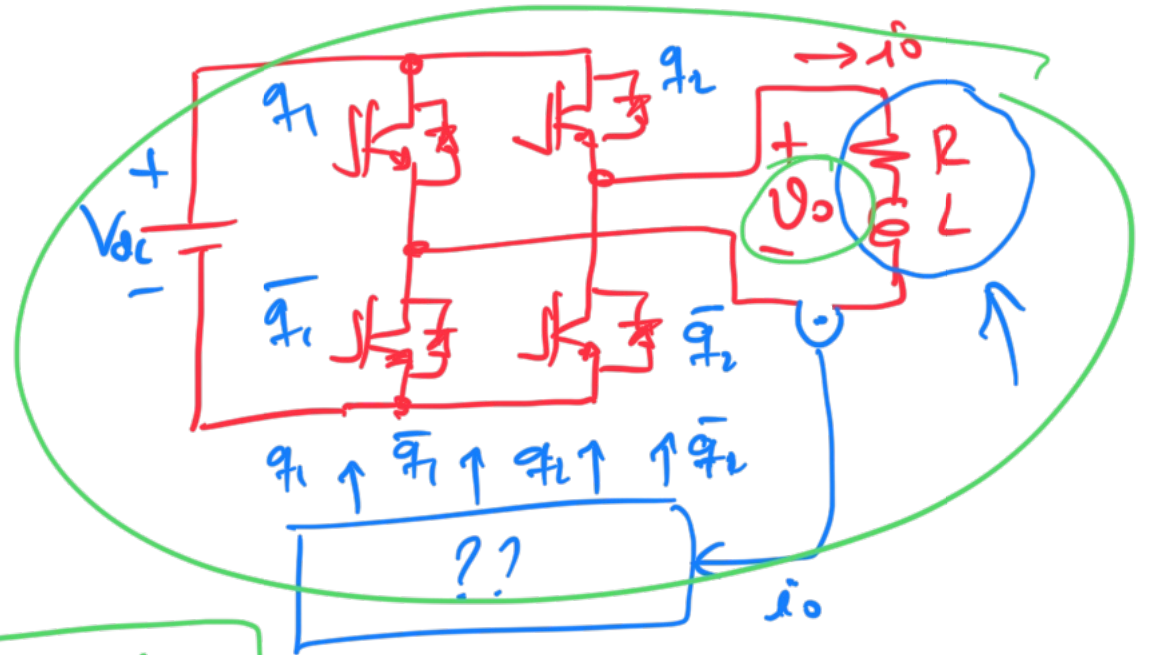
P2P

B2B CONF.



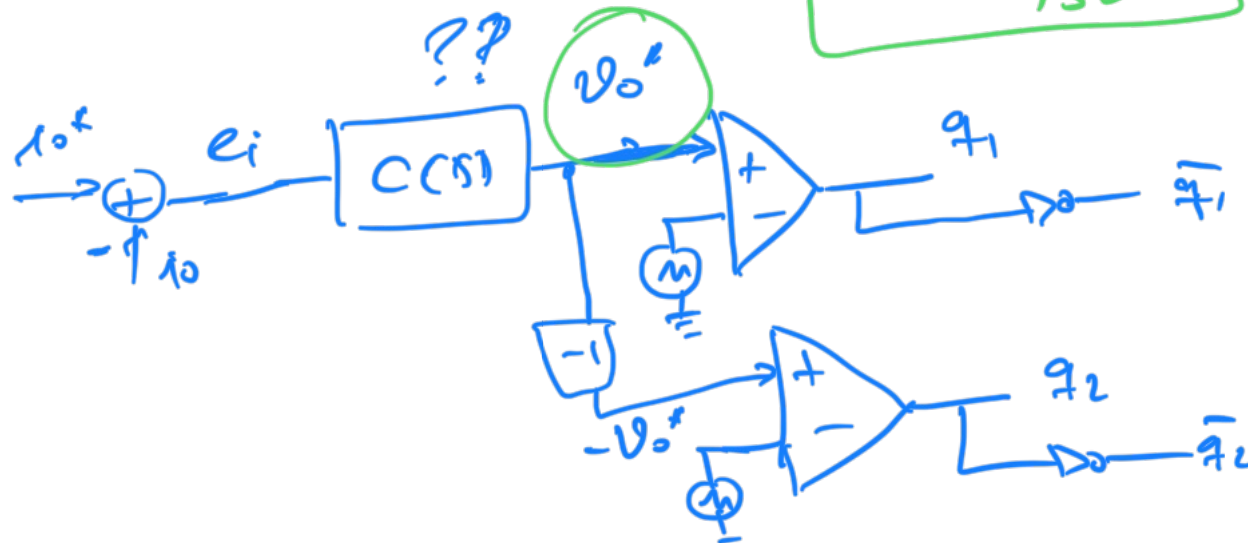


- $e_i \neq 0$ (-)
- $e_i = f(\Delta i, \omega_{AD}, V_{dc})$ (-)
- NO NEED TO DEFINE (+)



LINEAR CONTROL

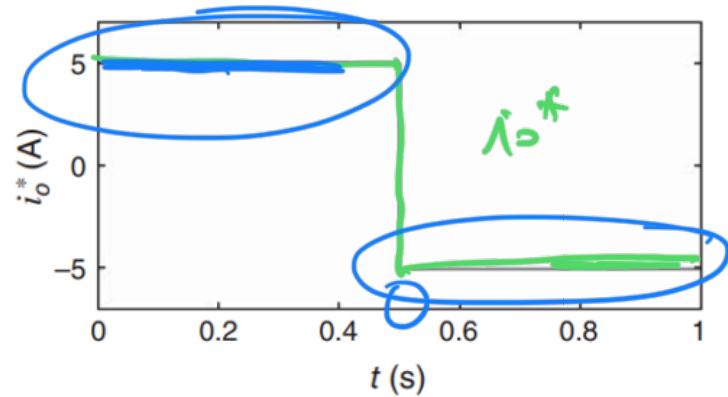
$$G(s) = \frac{1}{T_s s + 1}$$



$e_i = 0$ (+)
NEED TO DEFINE (-)

Problem 2

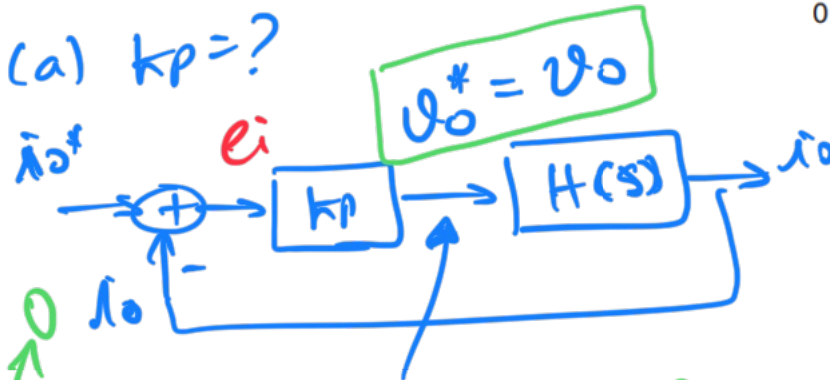
- (a) For Problem 1 with a proportional controller, determine the gain K_p that will lead to an error of steady state condition of 0.025 if the reference current is 5 A.
- (b) For the gain obtained in (a), determine the maximum and minimum voltage v_o applied to the load if the reference current is shown aside. (c) Repeat (b) if a half-bridge converter is used with dc-link voltage of 500 V.



$$R = 5 \Omega$$

$$L = 0.05 \text{ H}$$

$$i_o^* = 5 \text{ A} \quad (a) \quad k_p = ?$$



$$e_i = 0.025 \text{ A @ S.S.}$$

$$v_o = R i_o + L \frac{di_o}{dt} \quad (1)$$

$$e_i = i_o^* - i_o$$

$$0.025 = 5 - i_o$$

$$\therefore i_o = 5 - 0.025 = \underline{4.975 \text{ A}}$$

From (1):

$$v_o = 5 \cdot (4.975)$$

$$\boxed{v_o = 24.875 \text{ V}}$$

$$v_o = k_p \cdot e_i$$

$$k_p = \frac{24.875}{0.025}$$

$$\boxed{k_p = 995}$$



(b) $v_{o, \max} = ?$

$$v_o = k_p \cdot e_i$$

$$v_{o, \max} = k_p \cdot e_{i, \max}$$

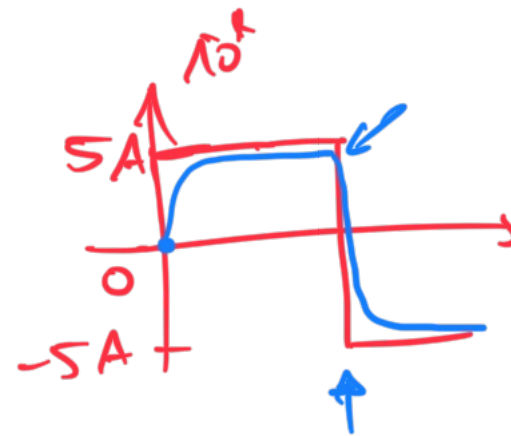
$$v_{o, \max} = 995 \cdot 5$$

$$v_{o, \max} = 4,975V$$

$$v_{o, \min} = k_p \cdot e_{i, \min}$$

$$v_{o, \min} = 995 \cdot [-5 - (5 - 0.025)]$$

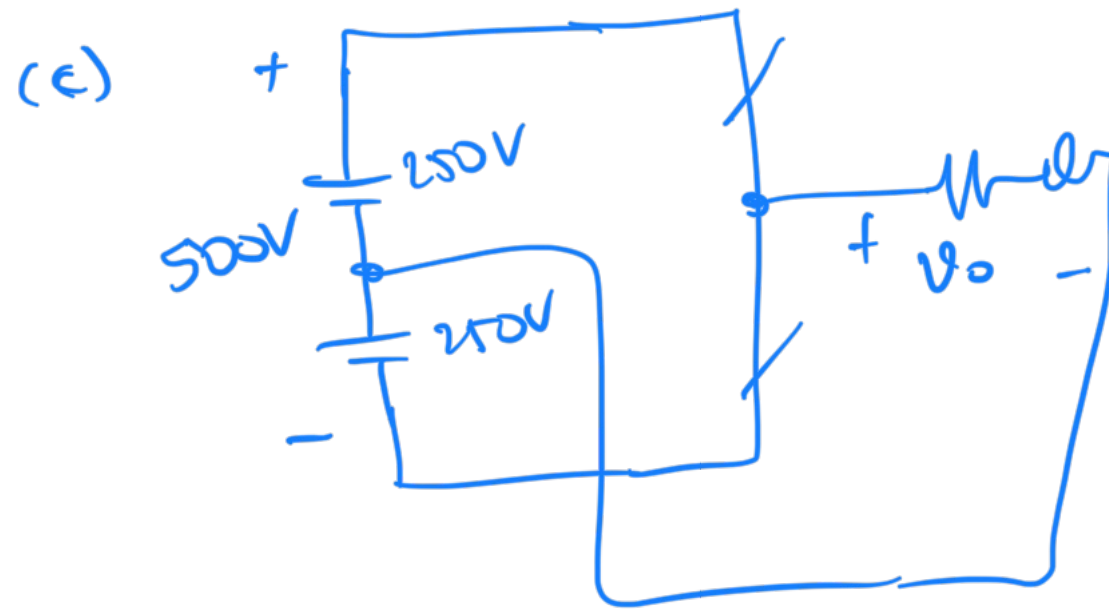
$$v_{o, \min} = 9,950V$$



$$e_{i, \max} = 10^4 - 10$$

$$= 5 - 0$$

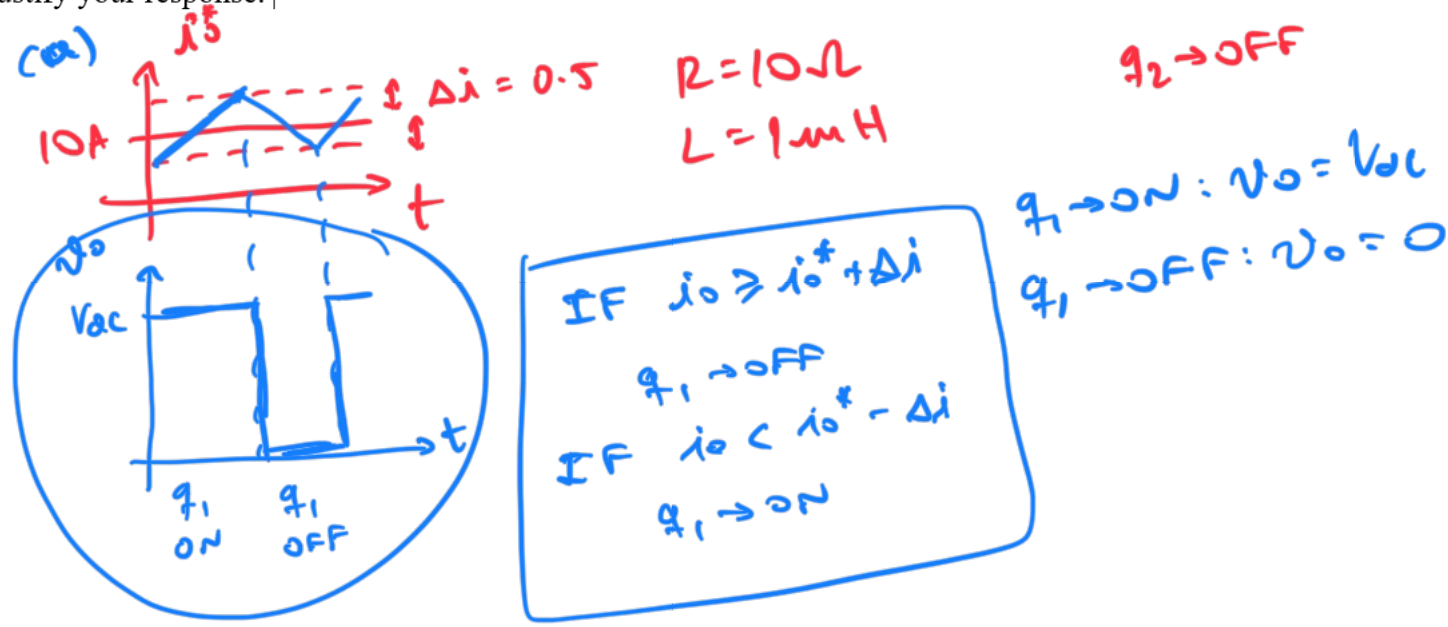
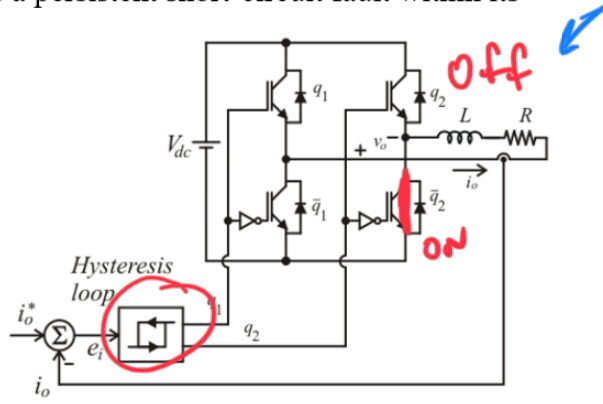
$$e_{i, \max} = 5A$$



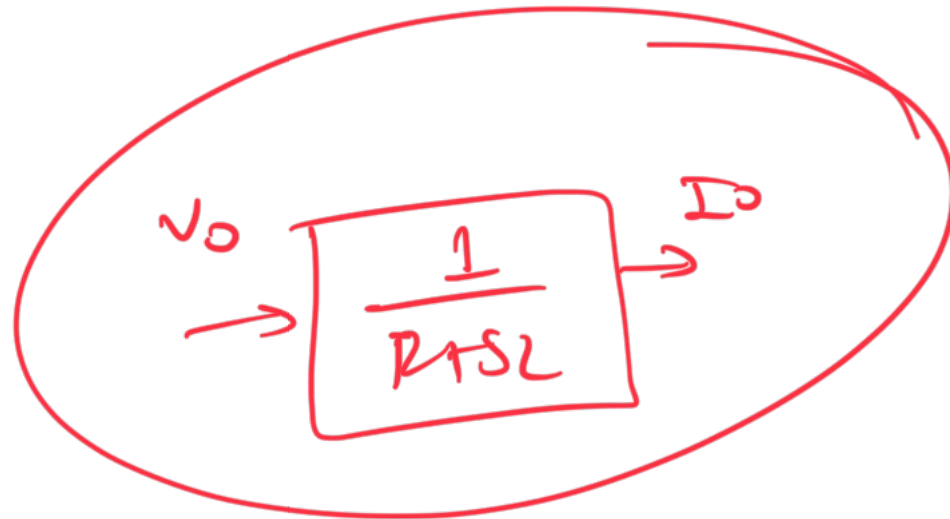
$$V_{o, \max} = 250V$$
$$V_{o, \min} = -250V$$

Problem 3

(a) The power electronic converter under investigation exhibits a persistent short-circuit fault within its switch $\bar{q}_2$, characterized by a continuous conductive state irrespective of the applied gate control signal. Change the hysteresis control logic of the other switches to guarantee the measured current follows a constant reference current equal to 10 A with ripple of 1 A. Assume $R = 10 \Omega$ and $L = 1 \text{ mH}$. (b) What's the minimum dc-link voltage to accomplish this control action. (c) Under fault conditions, can the current controller be implemented for this converter to track the reference current specified in Problem 2? Justify your response.



(b) $\langle i_o \rangle = 10\text{A}$
 $\langle v_o \rangle = 100\text{V}$ $V_{dc} \geq \underline{100\text{V}}$



Control Strategies for Power Converters

2. Application of the hysteresis control for dc motor drive

$$v_f = R_f i_f + L_f \frac{di_f}{dt}$$

$$v_a = R_a i_a + L_a \frac{di_a}{dt} + e_a$$

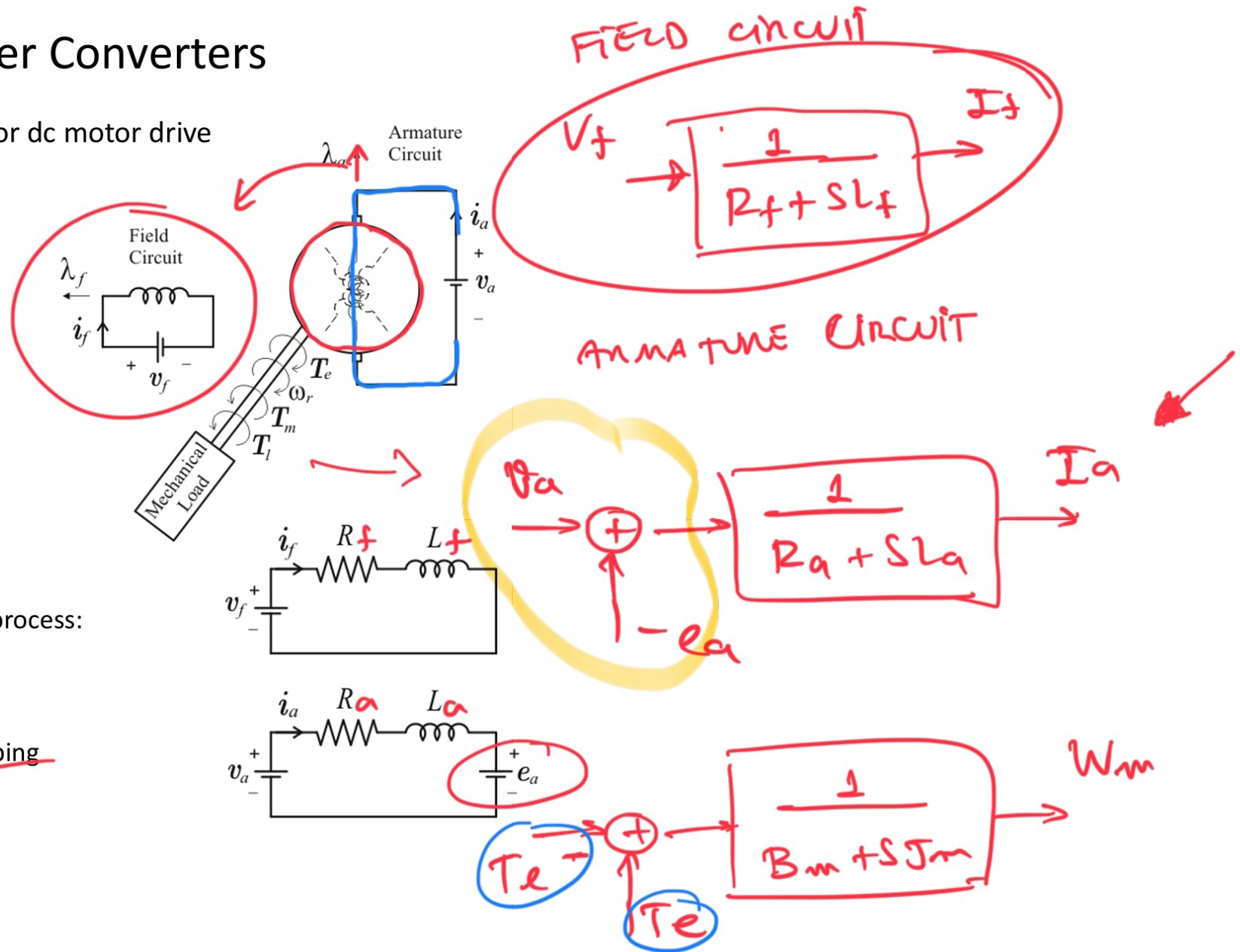
where $e_a = k_c \lambda_f \omega$ is the back emf (electromotive force); k_c is the coupling constant, λ_f is the field flux and ω is the mechanical speed

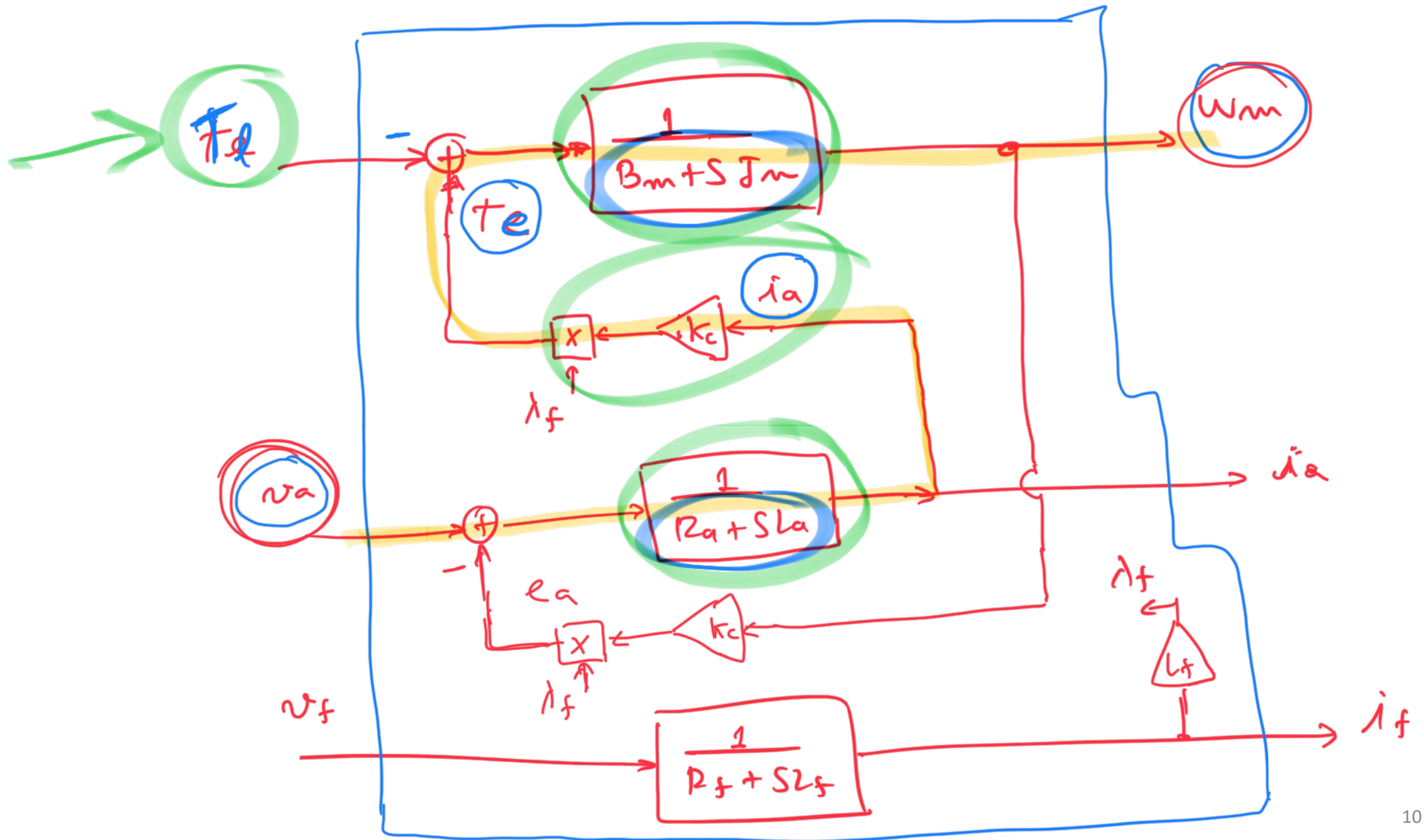
$$T_e = k_c \lambda_f i_a \quad (1)$$

Assuming no power loss in the conversion process:

$$T_e - T_l = B_m \omega + J_m \frac{d\omega}{dt}$$

where T_l is the load torque, B_m is the damping coefficient and J_m is the moment of inertia.



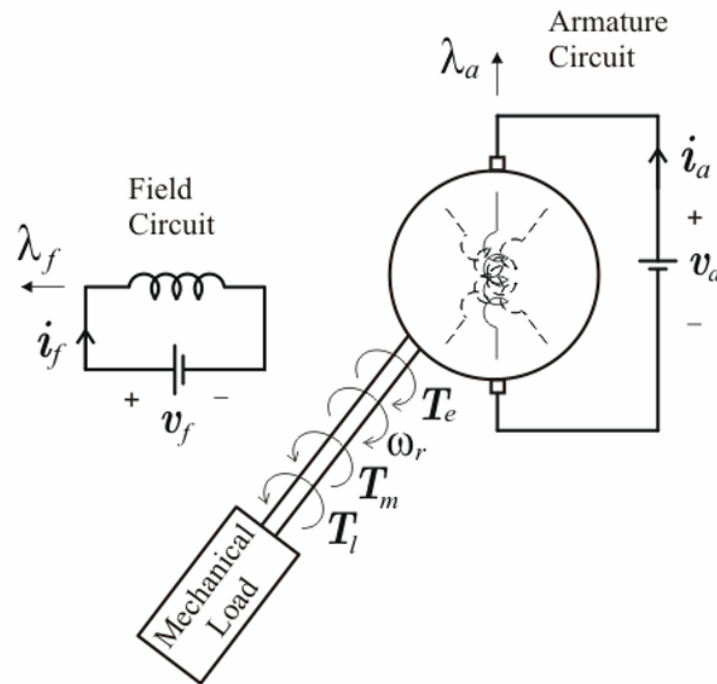


Control Strategies for Power Converters

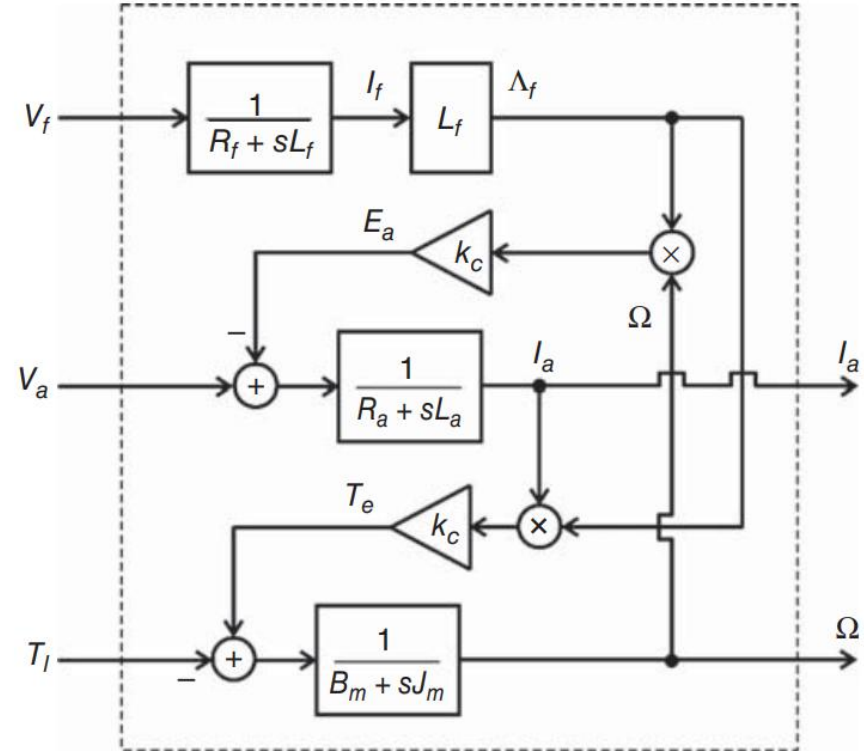
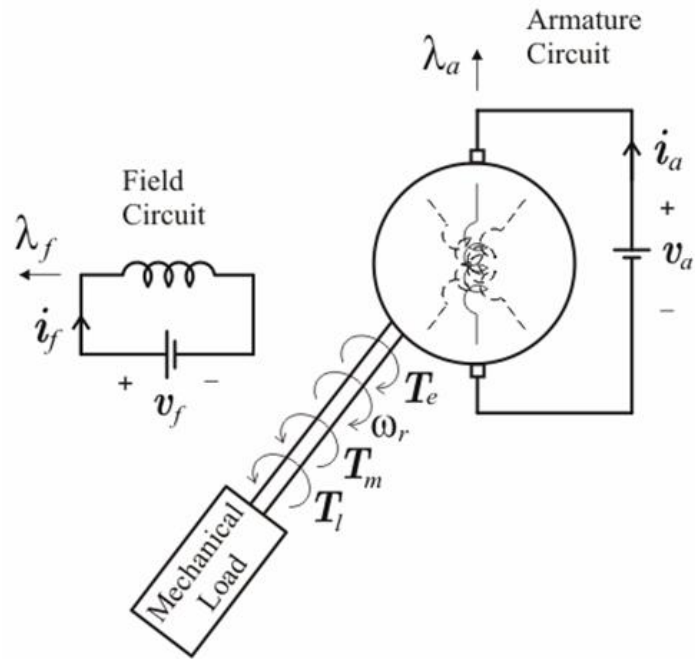
2. Application of the hysteresis control for dc motor drive

State space representation

$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$



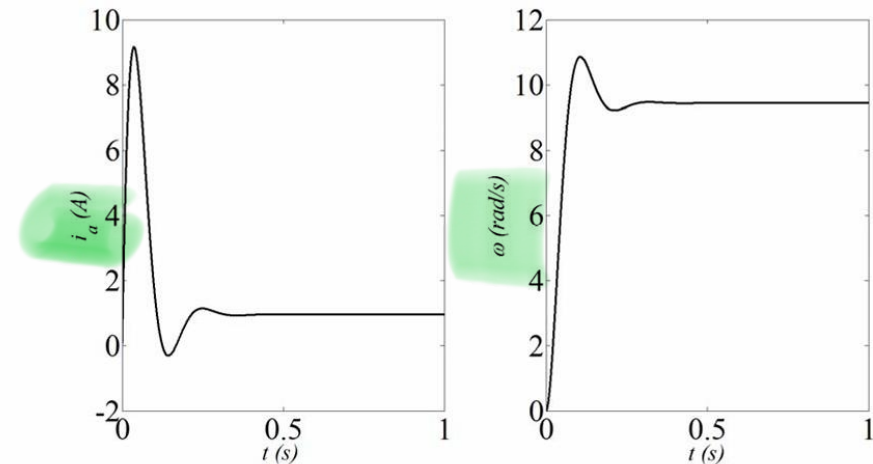
$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$



Control Strategies for Power Converters

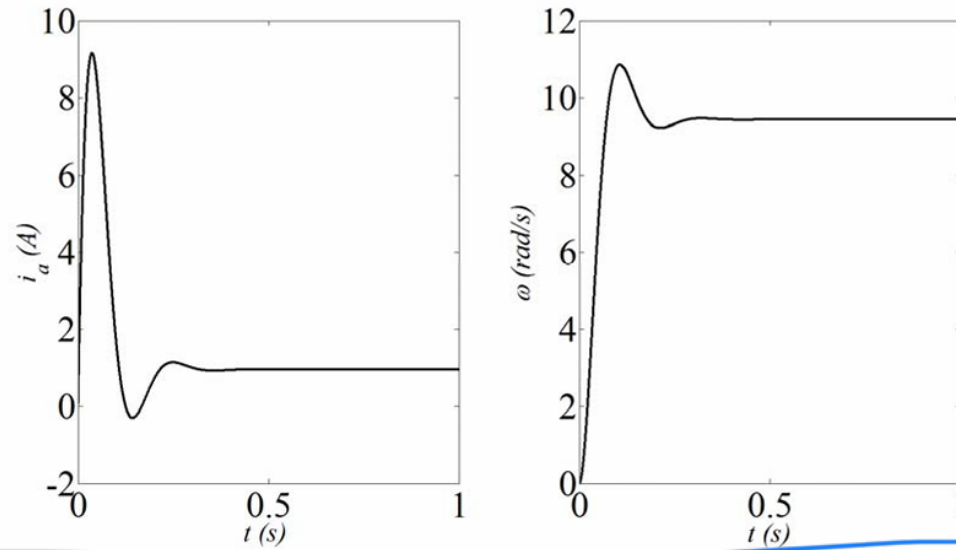
```
%constants
ra=0.06;
Laa=0.0018;
kf=0.1;
Bm=0.01;
Jm=0.001;
lambdaf=1;
%state variables
wr=0;
ia=0;
%input variables
va=1; %Armature voltage
Tl=.1; %Load torque
h=1e-4;%step size
j=0;
t=0;
tmax=2;
while t<=tmax,
t = t + h; % simulation time
ia=ia+(va/Laa-ra*ia/Laa-kf*lambdaf*wr/Laa)*h;%state
space 1
wr = wr + (kf*lambdaf*ia/Jm-Tl/Jm-Bm*wr/Jm)*h; %state
space 2
% storage the variables
j = j + 1;
time(j)=t;
armature_current(j)=ia;
rotor_speed(j)=wr;
end
figure(1),subplot(2,1,1),plot(time,armature_current)
subplot(2,1,2),plot(time,rotor_speed)
```

State space representation

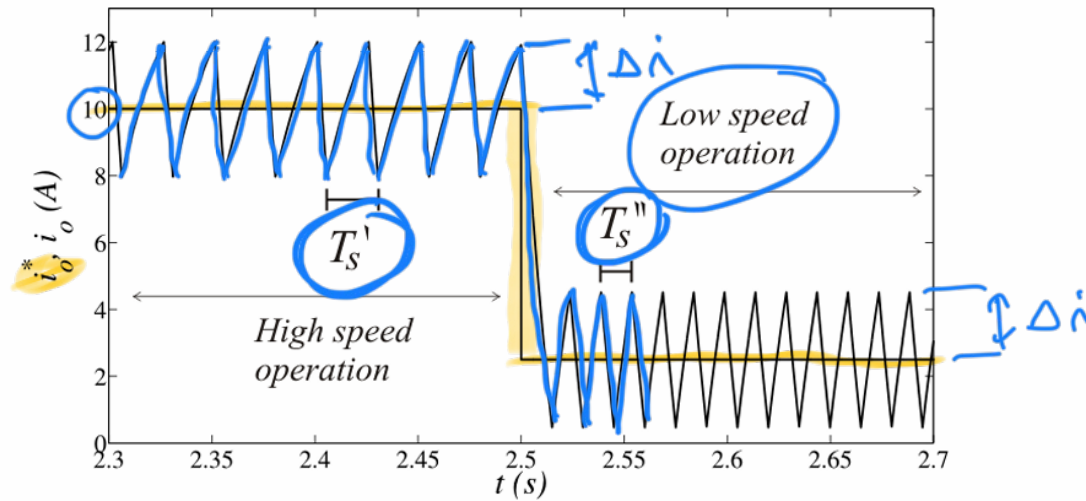
$$\frac{d}{dt} \begin{bmatrix} i_a \\ \omega \end{bmatrix} = \begin{bmatrix} -R_a/L_a & -k_c \lambda_f/L_a \\ k_c \lambda_f/J_m & -B_m/J_m \end{bmatrix} \begin{bmatrix} i_a \\ \omega \end{bmatrix} + \begin{bmatrix} 1/L_a & 0 \\ 0 & -1/J_m \end{bmatrix} \begin{bmatrix} v_a \\ T_l \end{bmatrix}$$


Control Strategies for Power Converters

Open-loop operation:



Close-loop operation:

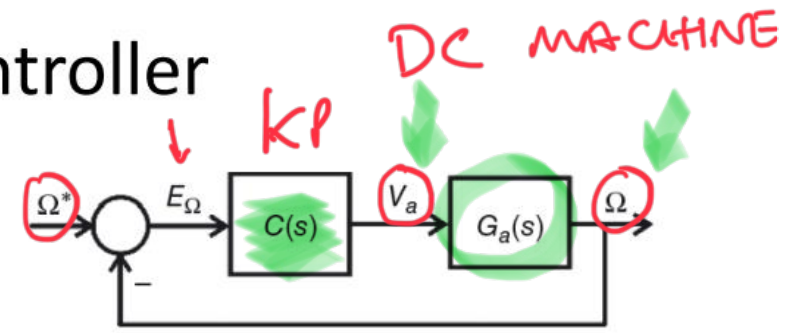
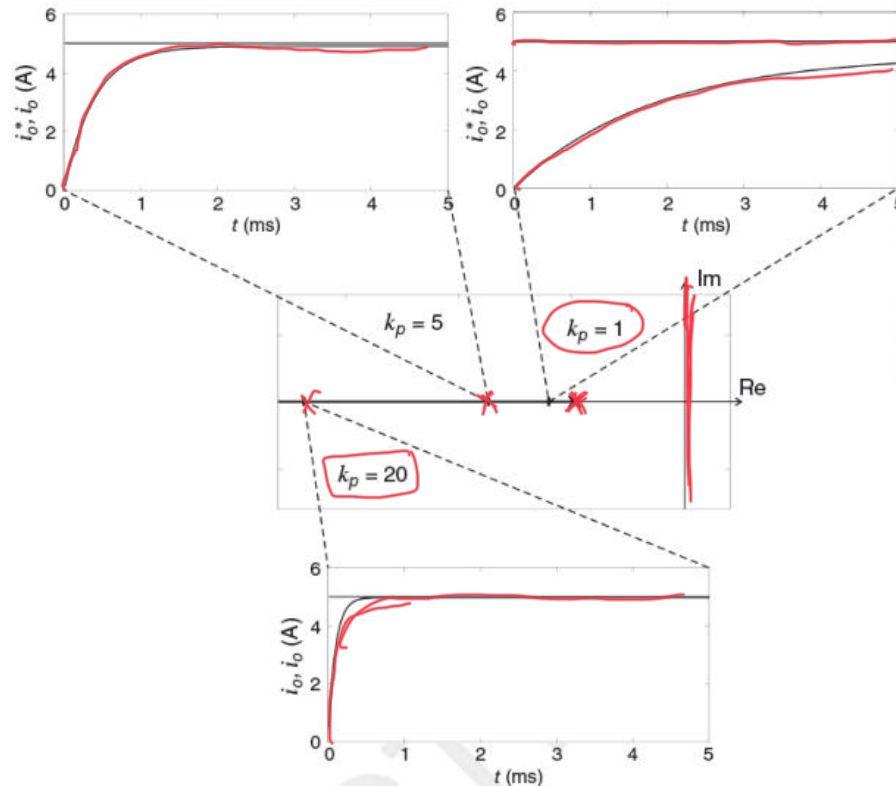


Root locus (RL)

Proportional Controller

$$F(s) = \frac{k_p}{R + sL} \quad \text{OLTF}$$

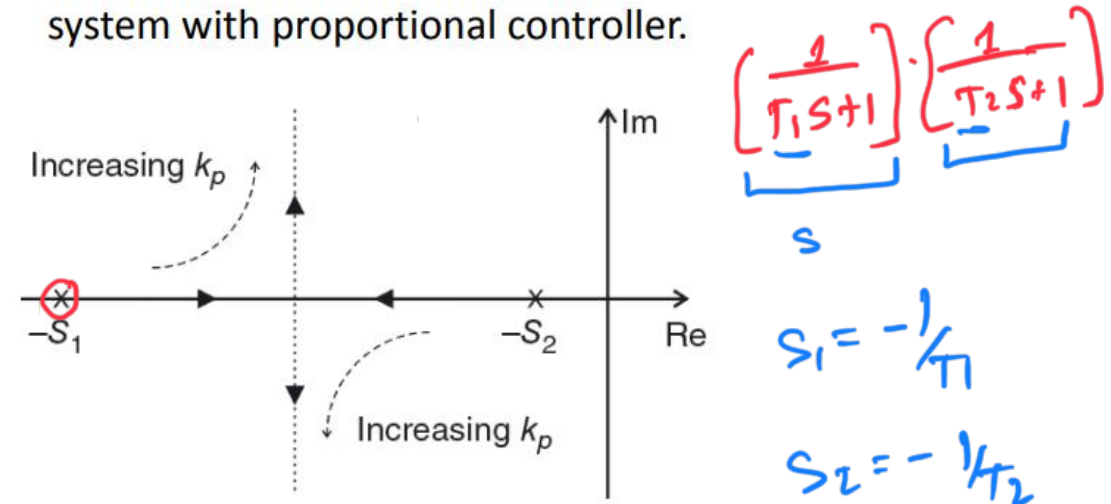
$$M(s) = \frac{k_p}{(R + k_p) + sL} \quad \text{CLTF}$$

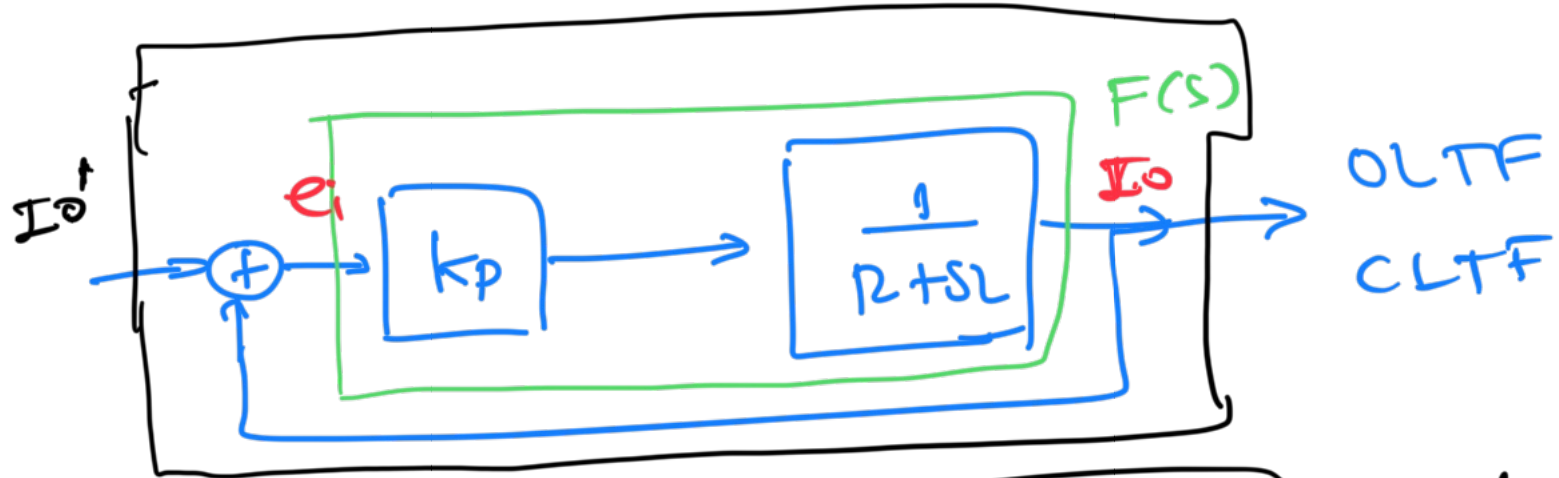


$$F(s) = \frac{k_p k_a}{(T_1 s + 1)(T_2 s + 1)} \quad \text{OLTF}$$

$$M(s) = \frac{k_p k_a}{(T_1 s + 1)(T_2 s + 1) + k_p k_a} \quad \text{CLTF}$$

Root locus of the dc motor drive system with proportional controller.





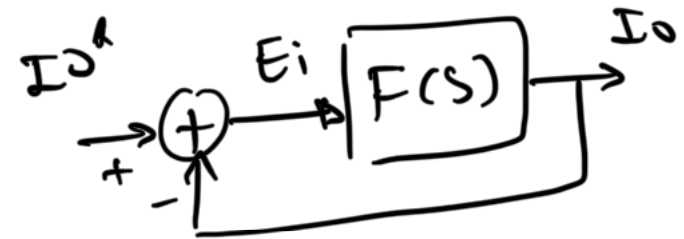
OLTF
CLTF

$$F(s) = \frac{I_o}{E_i} \quad \text{OLTF}$$

$$M(s) = \frac{I_o}{I_o^*} \quad \text{CLTF}$$

$$F(s) = k_p \cdot \frac{1}{R+sL}$$

$$M(s) = \frac{F(s)}{1+F(s)}$$



$$\frac{I_o}{I_o^*} = ?$$

$$E_i = I_o^* - I_o$$

$$I_o = F(s) \cdot (I_o^* - I_o)$$

$$I_o [1 + F(s)] = F(s) \cdot I_o^*$$

CHARACTERISTIC EQUATION:

$$1 + F(s) = 0 \Rightarrow 1 + \frac{k_p}{R+sL} = 0$$

RL

PI

Proportional-Integral Controller

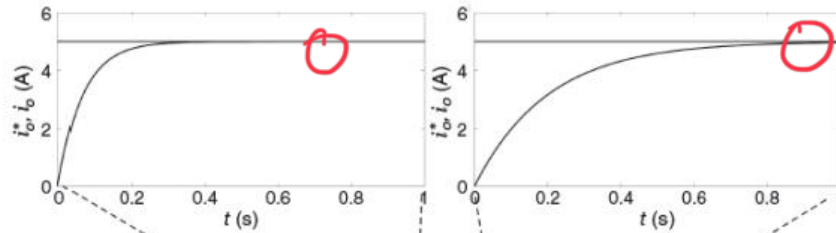
DC MOTOR

$$F(s) = \left(k_p + \frac{k_i}{s}\right) \left(\frac{1}{R + sL}\right) \quad \text{OLTF}$$

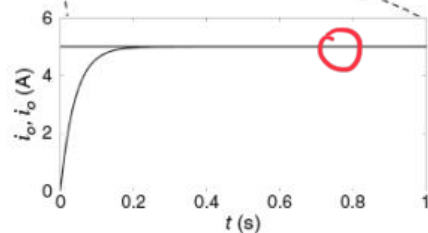
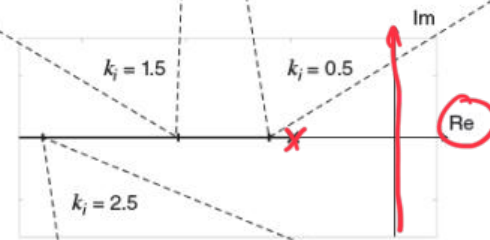
$$F(s) = \frac{k_i}{s} \left(\frac{k_p}{k_i} s + 1\right) \left(\frac{1/R}{\frac{L}{R}s + 1}\right) \Rightarrow \frac{k_i}{k_p} = \frac{R}{L}$$

$$F(s) = \frac{k_i}{sR}$$

$$M(s) = \frac{k_i}{sR + k_i}$$



$e_i = 0 \rightarrow \left[k_p + \left(\frac{k_i}{s}\right)\right] \rightarrow \omega_o \neq 0$

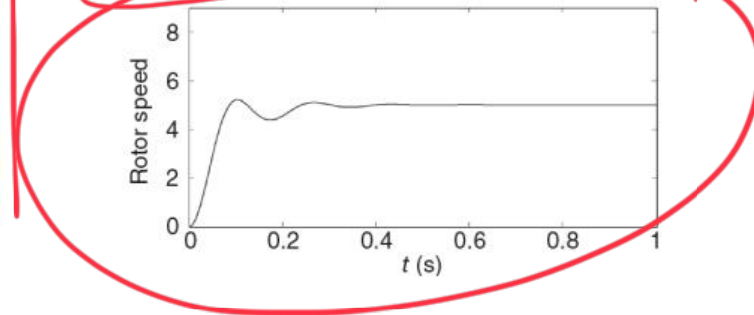
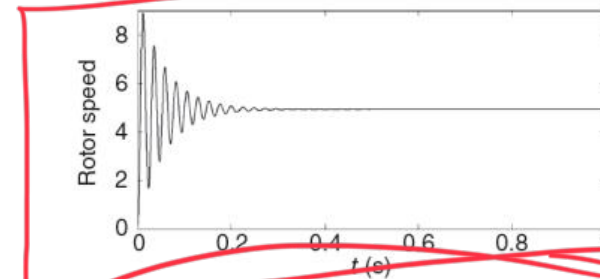


$$G_a(s) = \frac{k_a}{(T_1 s + 1)(T_2 s + 1)}$$

$$k_p/k_i = T_2$$

$$F(s) = \frac{k_i k_a}{s(T_1 s + 1)} \quad \text{OLTF}$$

$$M(s) = \frac{k_i k_a}{s(T_1 s + 1) + k_i k_a} \quad \text{CLTF}$$



Proportional-Integral-Derivative Controller

$$C(s) = k_p + \frac{k_i}{s} + k_d s$$

PID

$$C(s) = k_p + \frac{k_i}{s} + \frac{k_d s}{sT_d + 1}$$

←

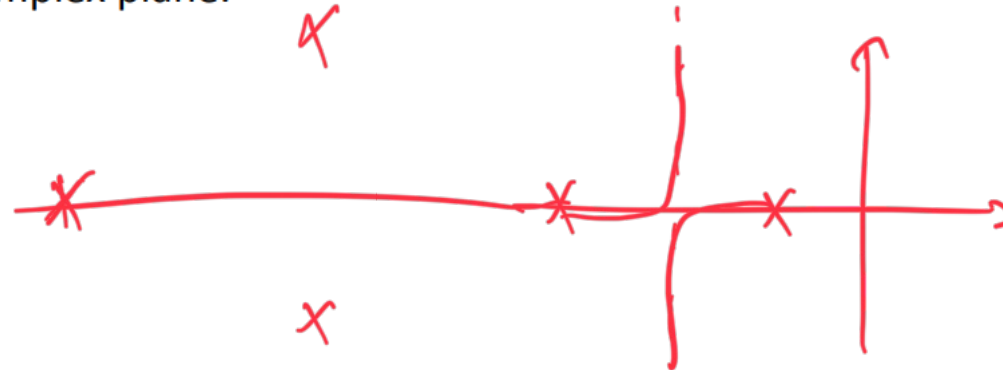
$$C(s) = \frac{(T_d k_p + k_d)s^2 + (T_d k_i + k_p)s + k_i}{s(T_d s + 1)}$$

$$k_i = \frac{1}{T_i} = \frac{1/k_a}{4T_d}$$

$$k_p = \frac{T_1 + T_2 - T_d}{T_i}$$

$$k_d = \frac{T_1 T_2 - (T_1 + T_2 - T_d)T_d}{T_i}$$

The parameters k_p and k_d can be employed to allocate the closed loop poles anywhere in the complex plane.



LINEAR CONTROL – ac VARIABLE

Resonant PI type of controller

$$C(s) = \frac{k_a s^2 + k_b s + k_c}{(s^2 + \omega^2)}$$

state space representation of this controller

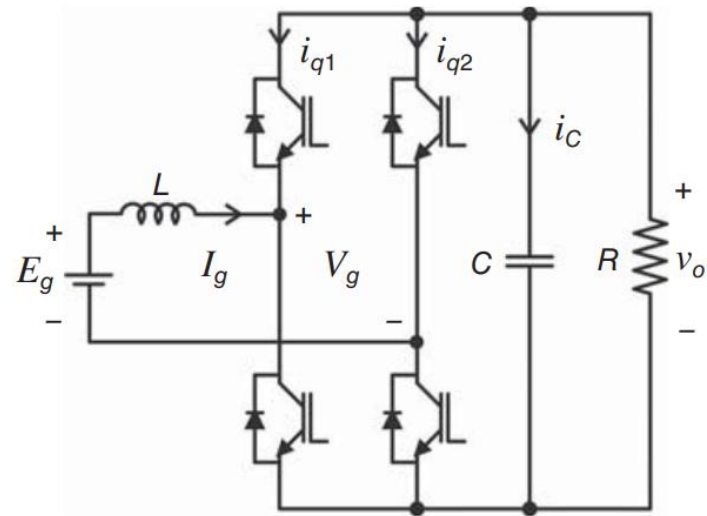
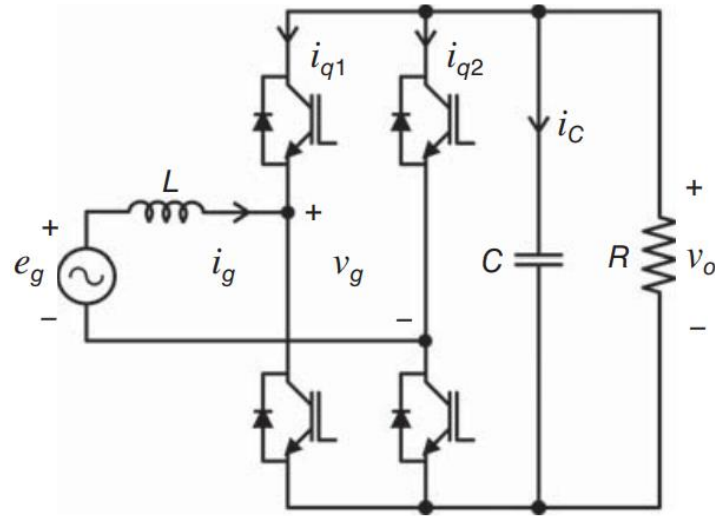
$$\dot{x}_1 = x_2 + k_b e$$

$$\dot{x}_2 = -\omega^2 x_1 + k_d e$$

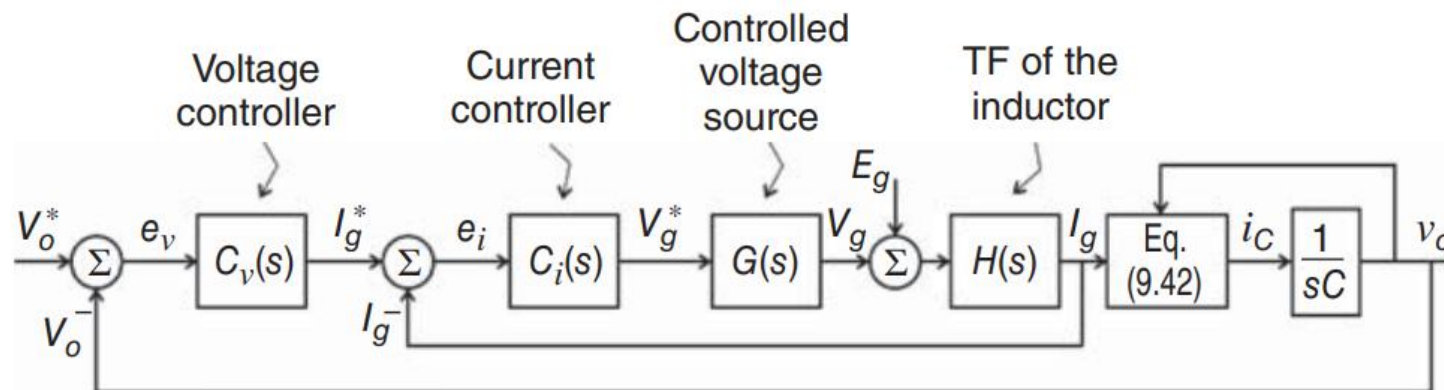
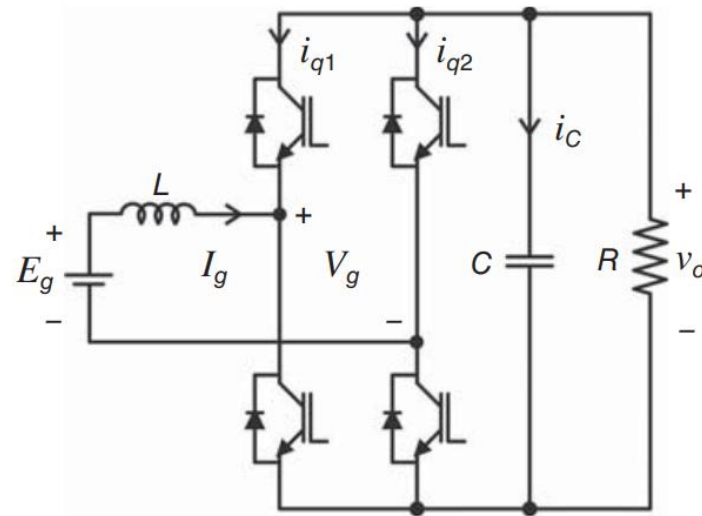
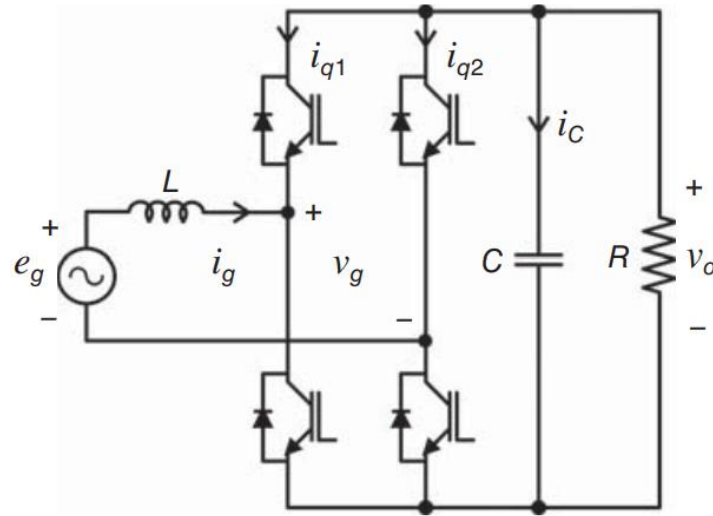
$$y = x_1 + k_a e$$

$$k_d = k_c - \omega^2 k_a$$

CASCADE CONTROL STRATEGIES



CASCADE CONTROL STRATEGIES



CASCADE CONTROL STRATEGIES

